



Multiple

Access

Techniques:

⇒

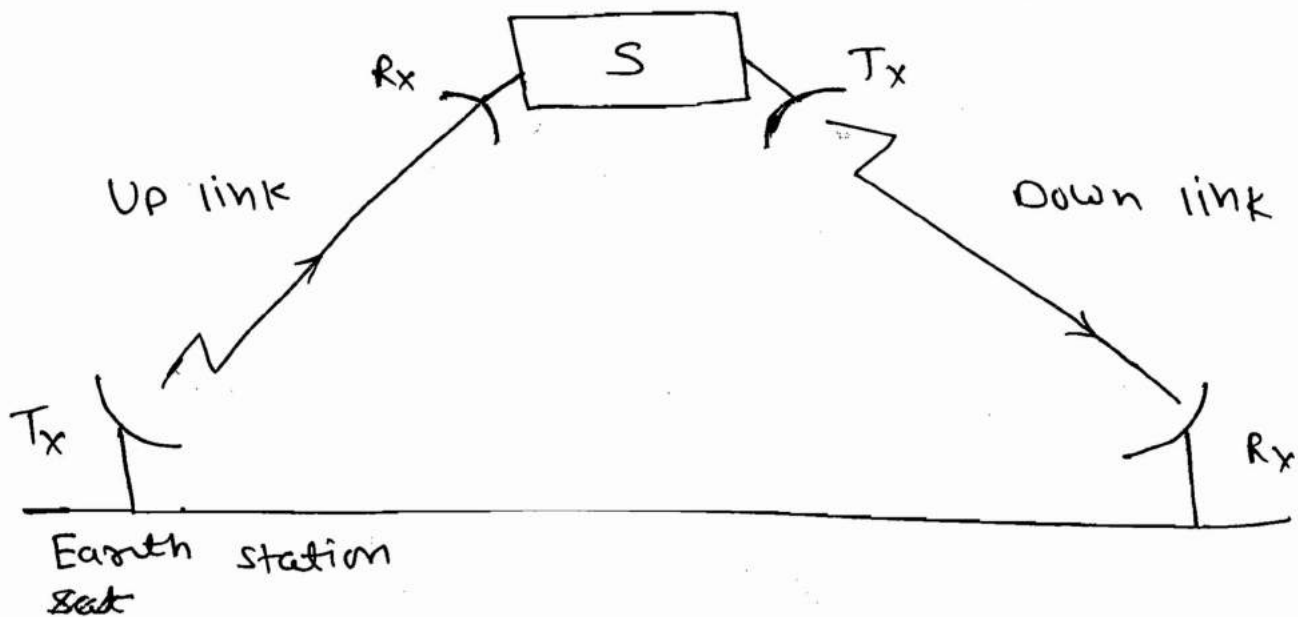
- ① FDMA
 - ② TDMA
 - ③ CDMA
- } Wireless Comm.

FDM } Wired Comm.
TDM }

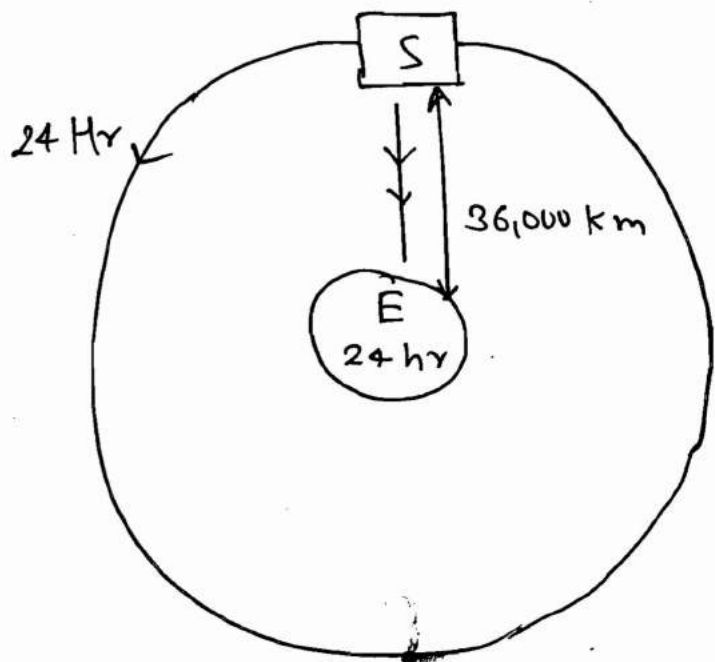
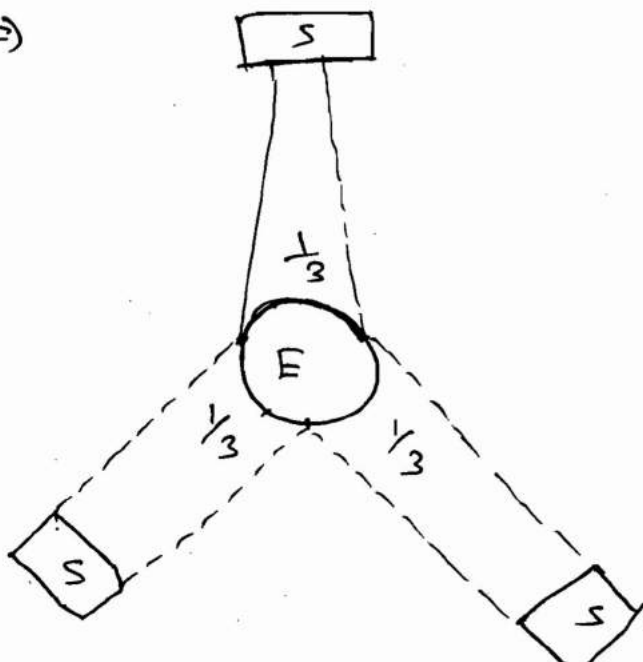
⇒ Consider the example of Satellite Communication to understand the above techniques.

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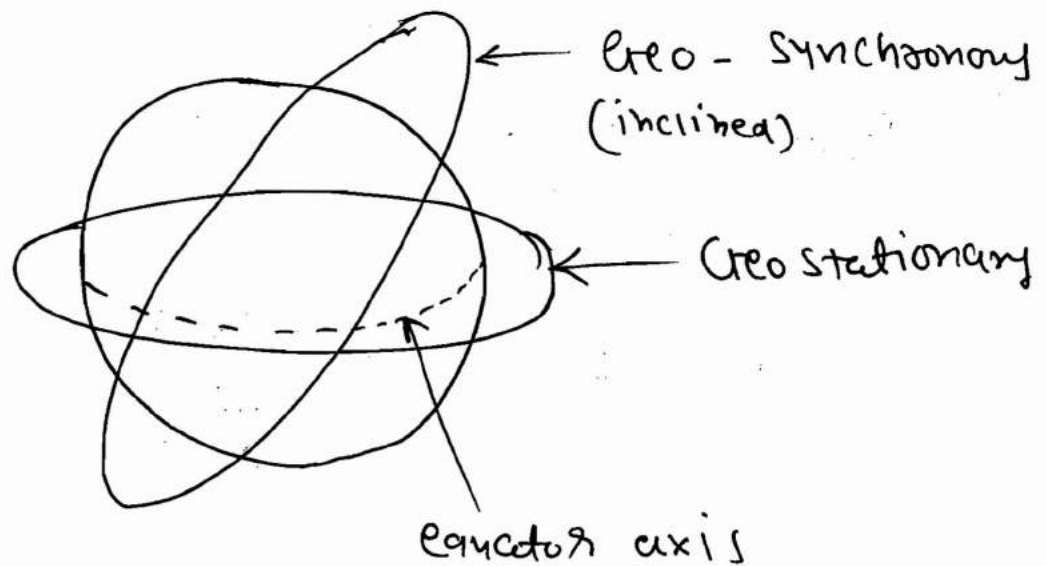
Microwave Repeater (GHz)



⇒



⇒ Geo-stationary and geo-synchronous:



⇒ Throughout the world, 3 combinations for Uplink & Down Link are used:

6/4 GHz.

14/12 GHz.

30/20 GHz.

Q Why Downlink is not same as UPLINK?

⇒ If Downlink is same as the Uplink, then interference occurs. If

UL = DL = 6 GHz.

⇒ Then, when R_x of satellite transmit the data it may be received by its own Receiver. So, interference will be occurred.

Q Why Uplink should be greater than the Downlink?

Ans: Power received by the Rx Es is given by,

$$P_r = 10 \log \left[\frac{P_t \cdot G_t \cdot G_r}{\left(\frac{4\pi R}{\lambda} \right)^2} \right]$$

$$\Rightarrow P_r = 10 \log [P_t G_t] + 10 \log [G_r] - \underbrace{10 \log \left[\left(\frac{4\pi R}{\lambda} \right)^2 \right]}_{\text{Path loss}}$$

⇒ As the Satellite produces power from the solar array, power produce is limited. one way to increase received power is by reducing the path loss. In order to reduce the path loss, λ should be increase and as $\lambda = \frac{c}{f}$, freq. should be decrease. That's why down link should be less than the up link.

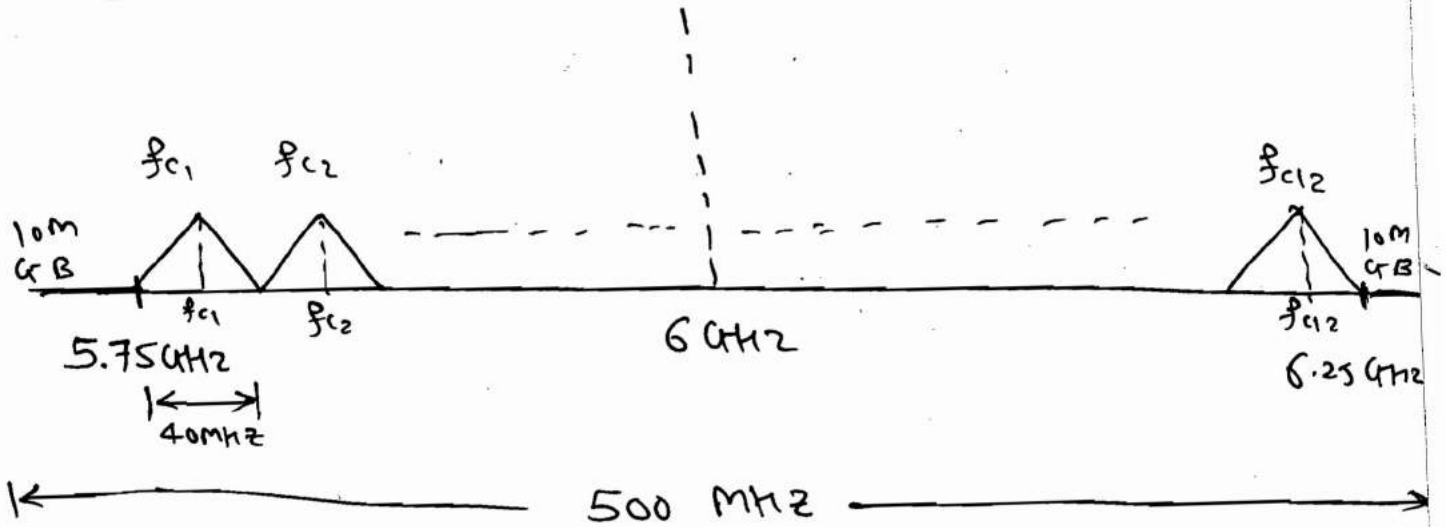
* Disadvantages:

- ① Very large loss as $R = 36,000$ km.
- ② Delay betⁿ T_x & R_x signal as $R = 36,000$ km

① FDMA: (Frequency Division Multiple Access).

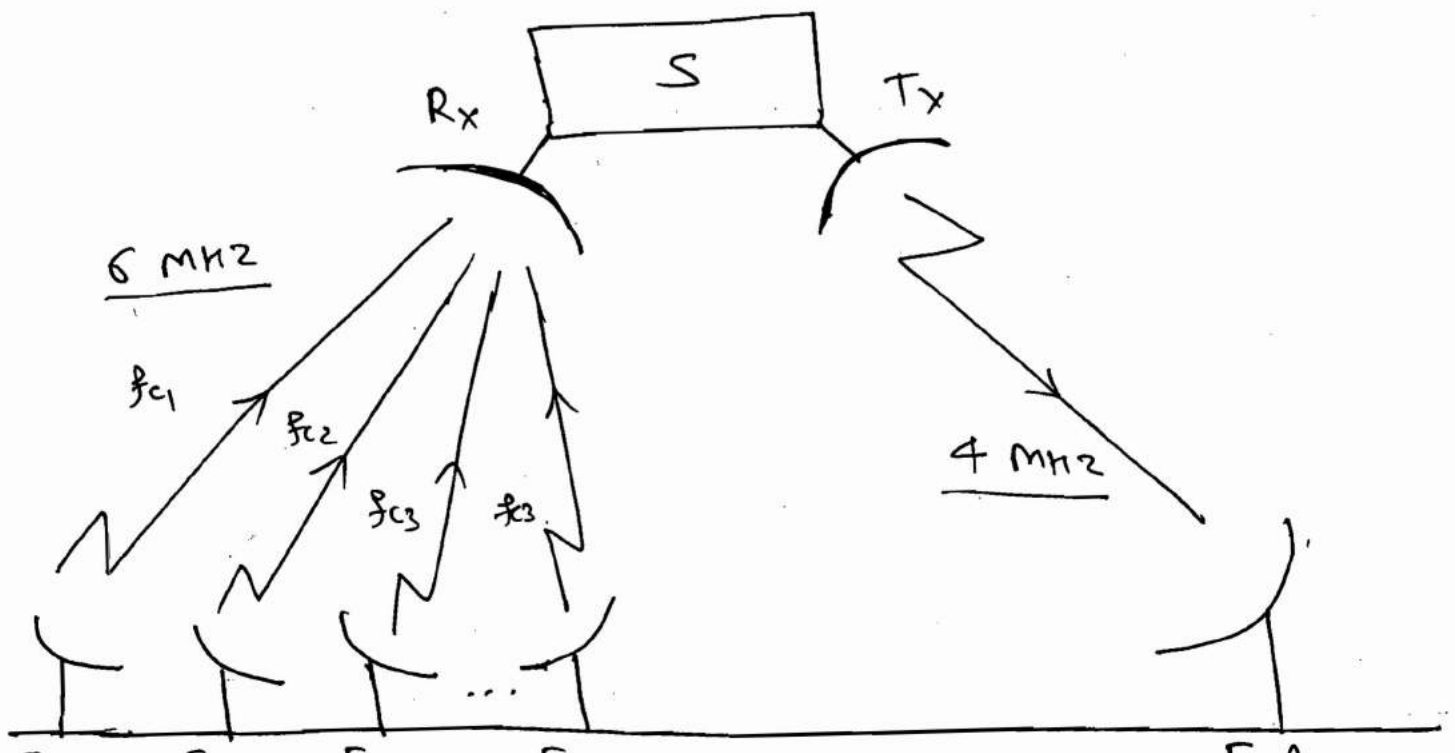
⇒ In FDMA, Multiple users will access the satellite with the allotted carrier freq.

⇒ BW = 500 MHz.

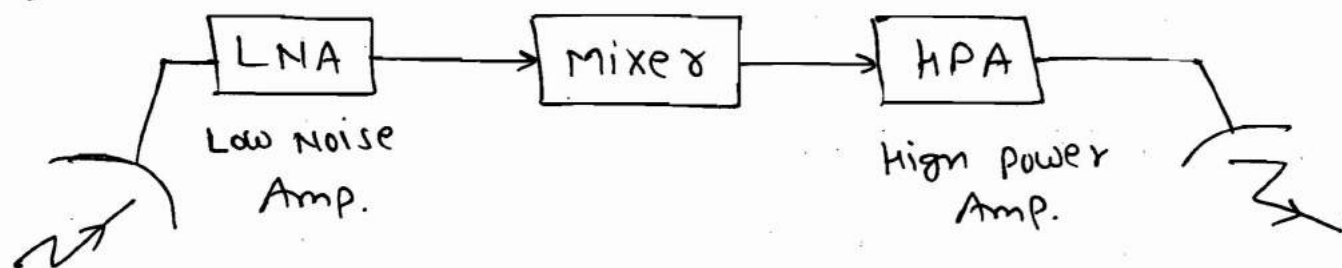


⇒ 1st satellite INTELSAT-1 by USA.

⇒ India 1st satellite INSAT-1

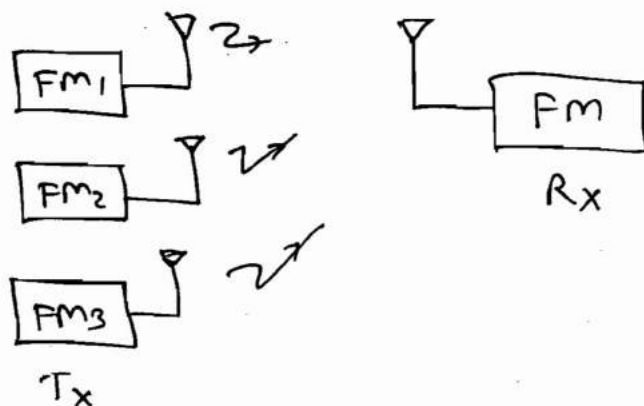


⇒



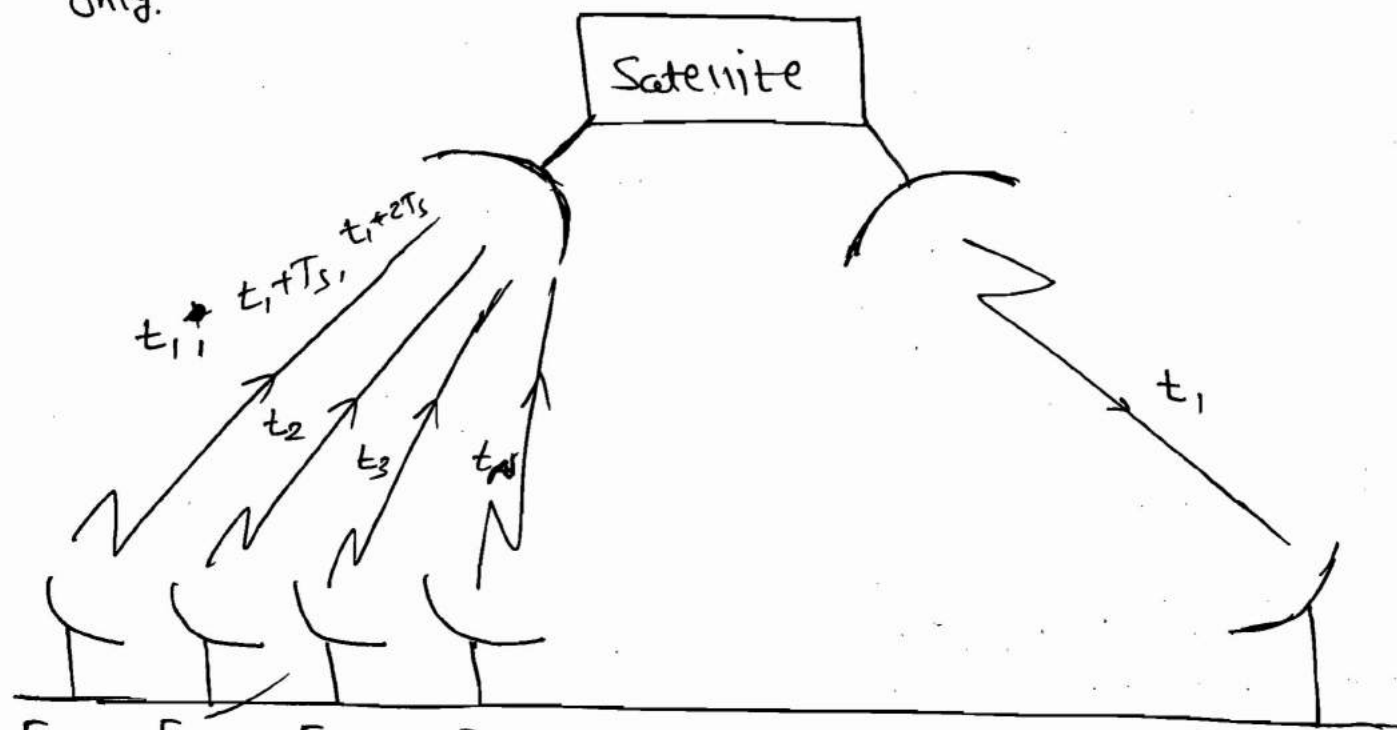
Transponder.

⇒ Another example is: Radio

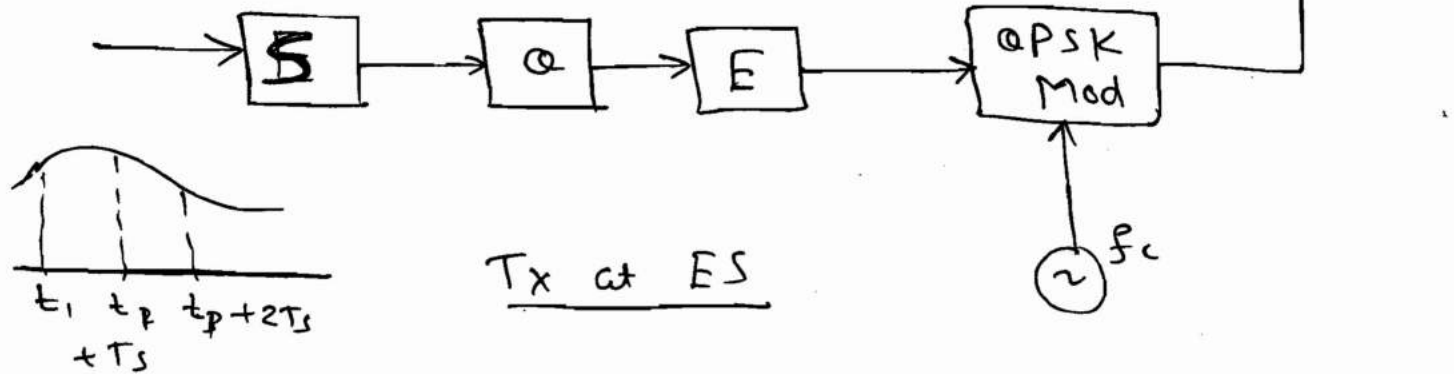


② TDMA: (Time Division Multiple Access).

⇒ In TDMA multiple users will access the satellite in the allotted time slot only.

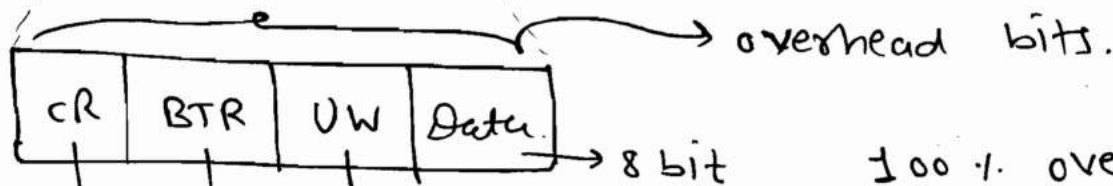
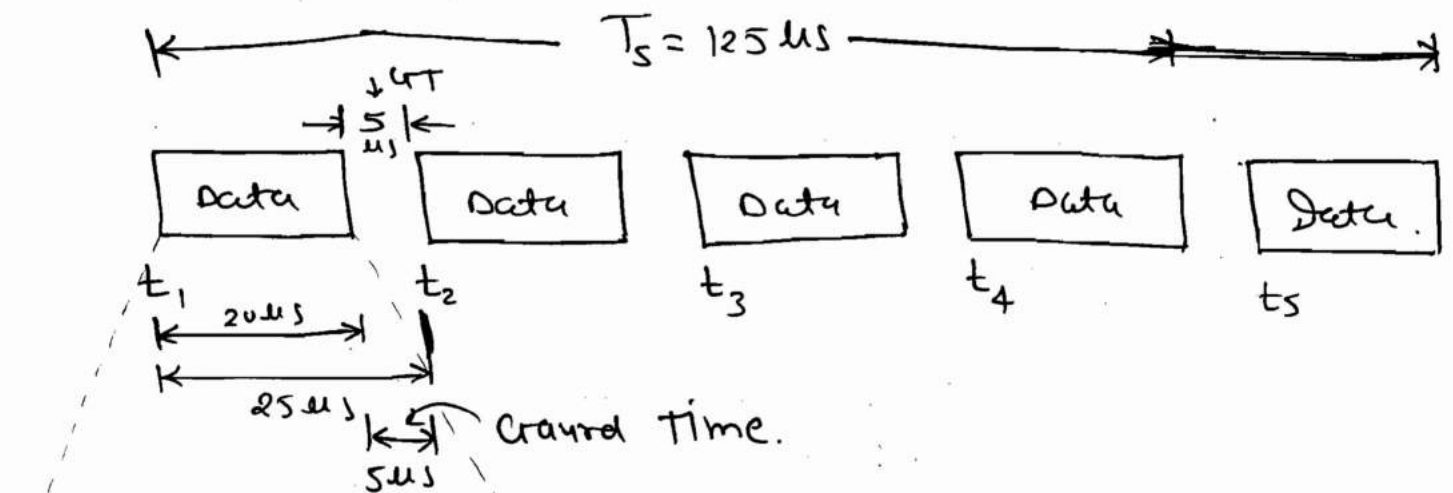


$\Rightarrow$ Voice, $\frac{1}{T_s} = 8000$



$\Rightarrow \frac{1}{T_s} = 8000 \Rightarrow T_s = 125 \mu s.$

$\Rightarrow$ TDMA Frame:



Carrier Recovery
Bit timing Recovery
Unique Word

means
$$\text{bits} (CR + BTR + UW) = (\text{Data bits}).$$

in this case

Data bits = 8

So, 100% overhead bits = 8.

Procedure:

① $T_s = ?$

$$T_s = 125 \text{ } \mu\text{s}.$$

② No. of ES = 5.

③ Time slot to each earth station

$$T_1 = \frac{125 \text{ } \mu\text{s}}{5} = 25 \text{ } \mu\text{sec} = \frac{\textcircled{1}}{\textcircled{2}}.$$

④ $T_1 - GT = 25 - 5 = 20 \text{ } \mu\text{sec}.$

$$\text{as } GT = 5 \text{ } \mu\text{sec}.$$

⑤ No. of overhead bits + data bits

$$= 12 + 8 = 20 \text{ bits}.$$

⑥ $T_b = \frac{20 \text{ } \mu\text{sec}}{20 \text{ bits}} = 1 \text{ } \mu\text{sec/bit} = \frac{\textcircled{4}}{\textcircled{5}}$

⑦ $R_b = \frac{1}{T_b}$

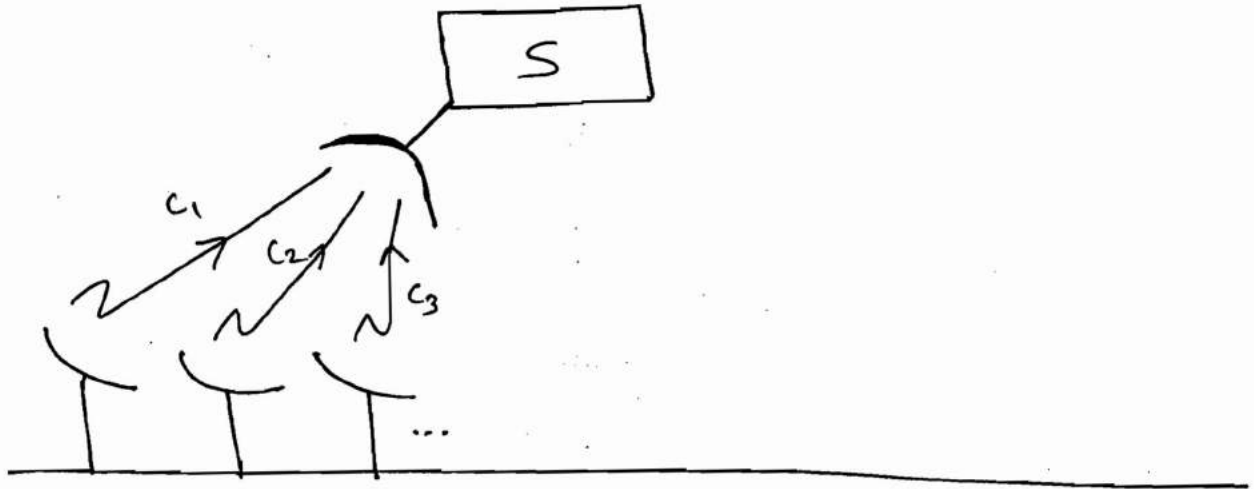
$$\Rightarrow R_b = \frac{1}{1 \text{ } \mu\text{sec/bit}}$$

$$\Rightarrow \boxed{R_b = 1 \text{ Mbps}}$$

③ CDMA: (Code Division Multiple Access):

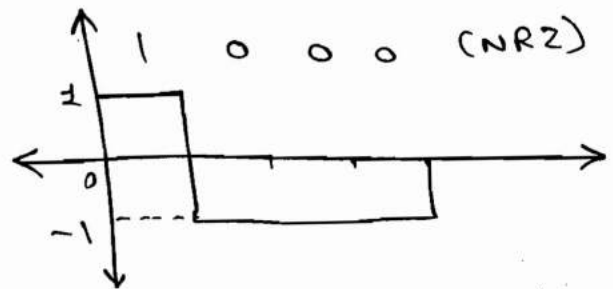
⇒ In CDMA, All the users can access the satellite at the same time and all signals will occupy the same freq. range.

⇒



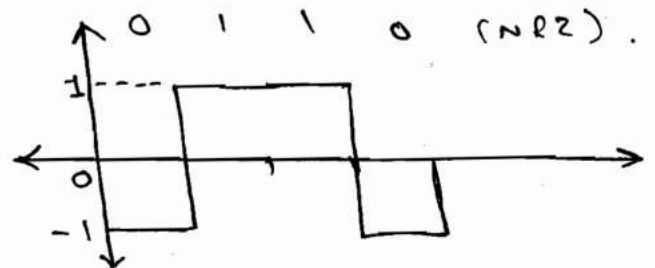
→

$$C_1 = 1 \ 0 \ 0 \ 0$$



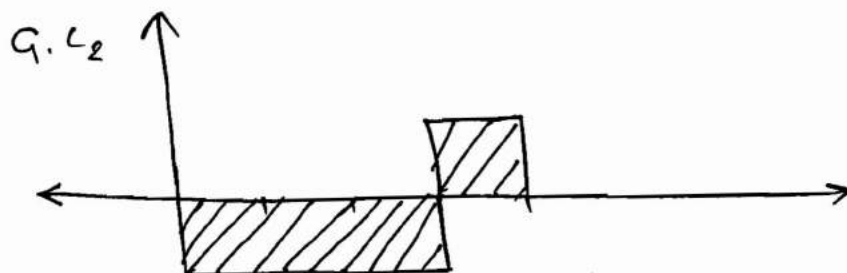
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$$C_2 = 0 \ 1 \ 1 \ 0$$



$$\Rightarrow \int_0^{4T_b} f_1(t) \cdot f_2(t) dt \neq 0.$$

So, not orthogonal to each other.

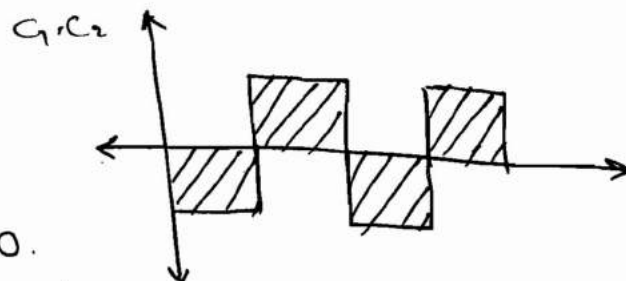


$$\Rightarrow \text{if } C_1 = 1000$$

$$C_2 = 0010$$

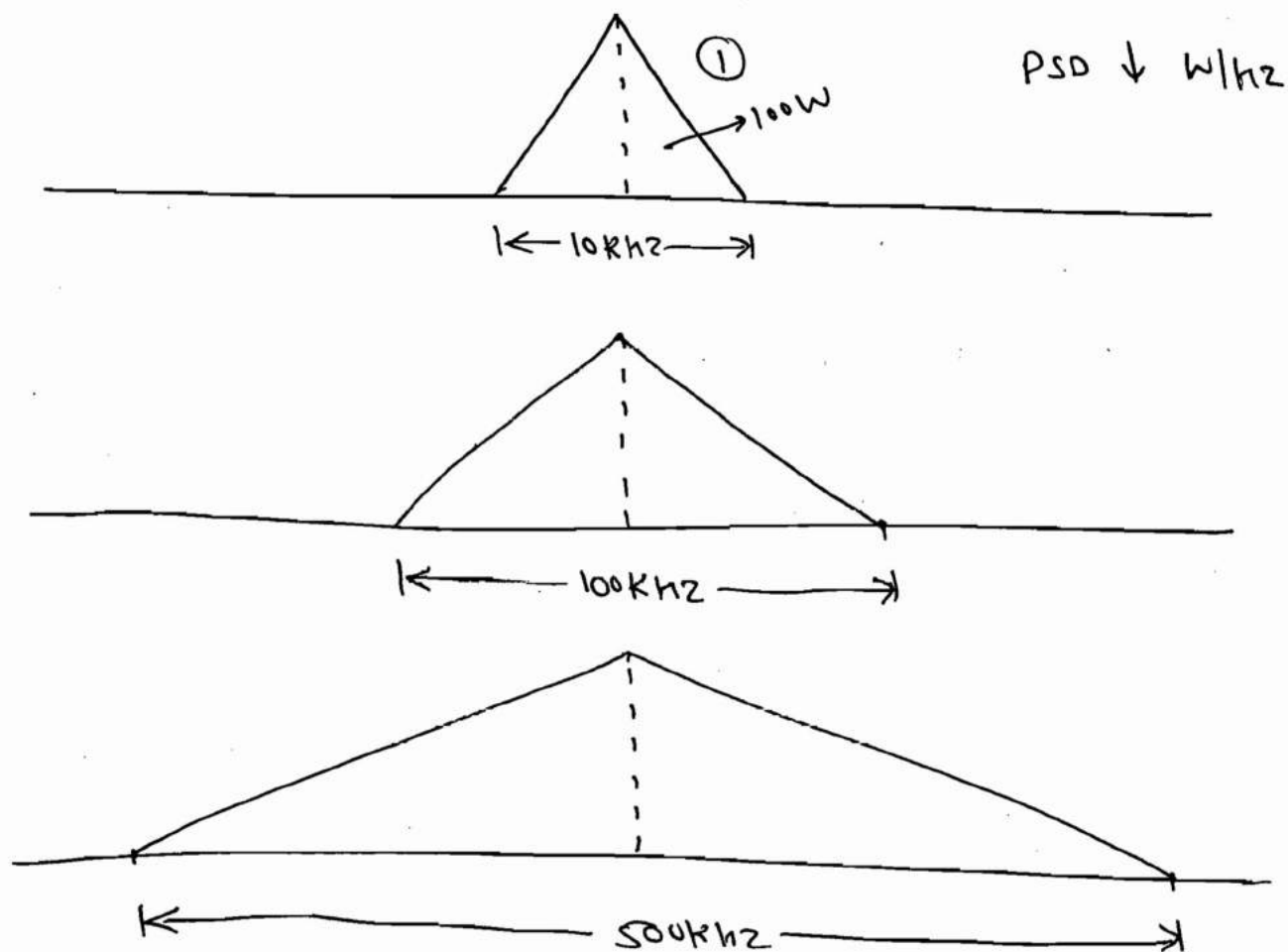
then

$$\int_0^{4T_b} f_1(t) \cdot f_2(t) dt = 0.$$



* Spread Spectrum modulation:

$\Rightarrow$



$\Rightarrow$ PN Sequence Code:

Property: ① All codes are orthogonal to each other.

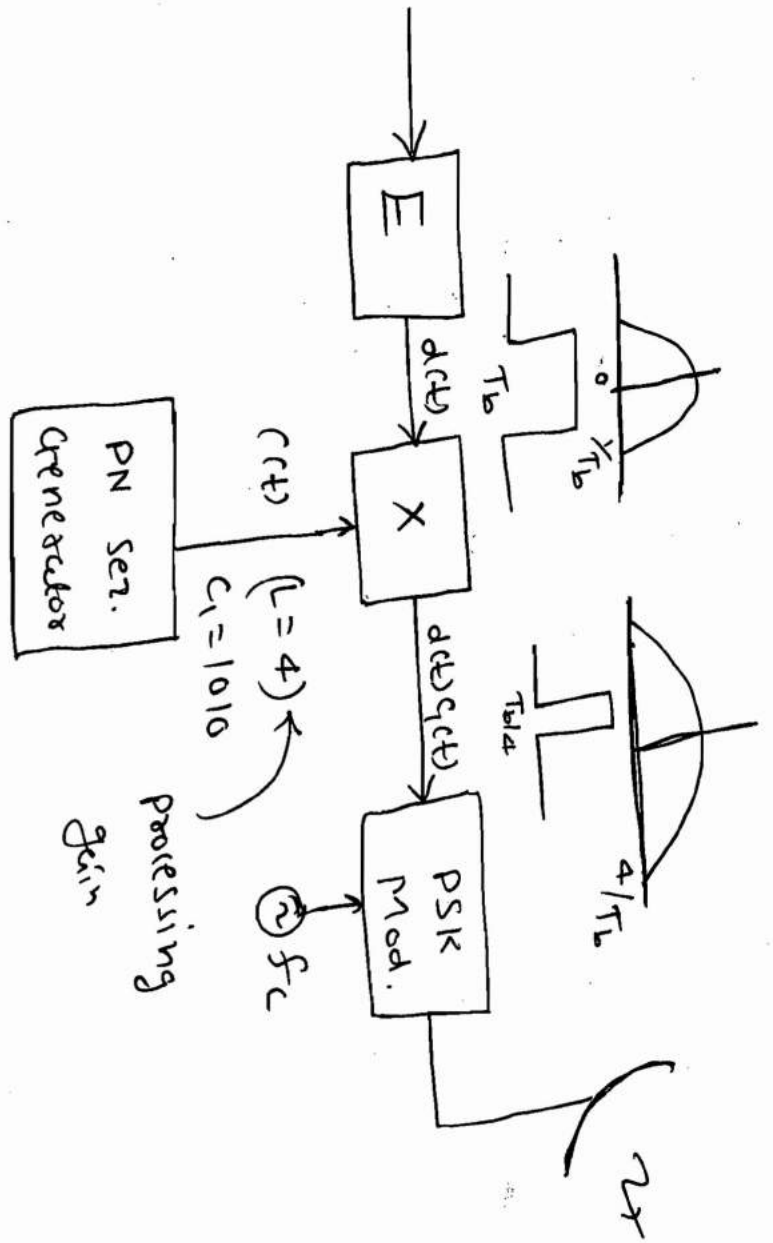
$$② \quad C_1^2(t) = 1$$

$$C_2^2(t) = 1$$

$$C_3^2(t) = 1$$

* CDMA Block Diagram:

ES Tx



ES Rx

